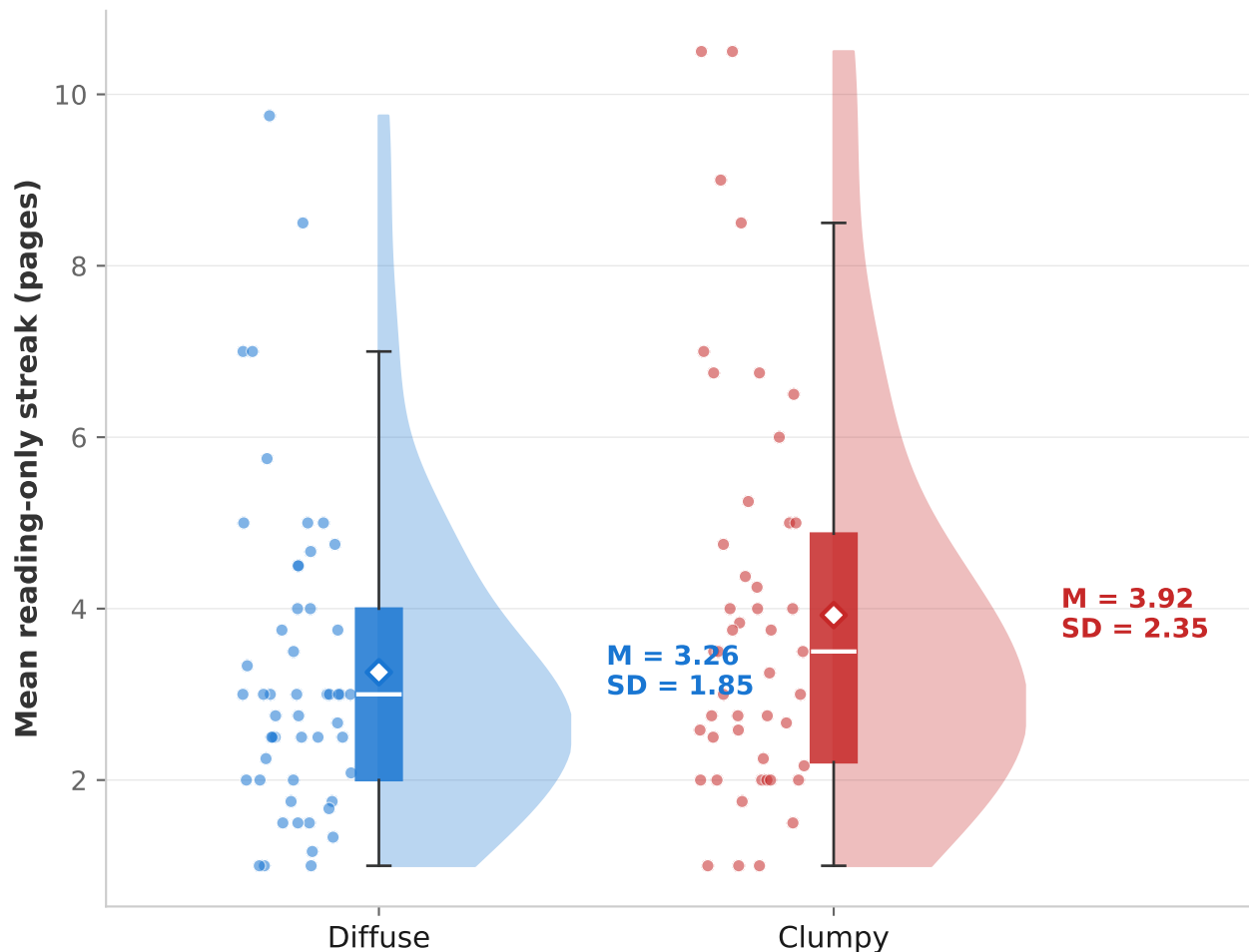


Mean reading-only streak: distribution shape by condition

Per participant (N = 99 of 101). The Clumpy condition opens the upper tail - not just shifts the mean.



Equality-of-variance tests

SD Diffuse 1.85

SD Clumpy 2.35

Variance ratio C/D 1.62

Levene (mean) $p = 0.139$

Brown-Forsythe $p = 0.158$

F-test $p = 0.096$

Variance gap is suggestive but not formally significant (N = 99 - underpowered for a 27% SD difference).

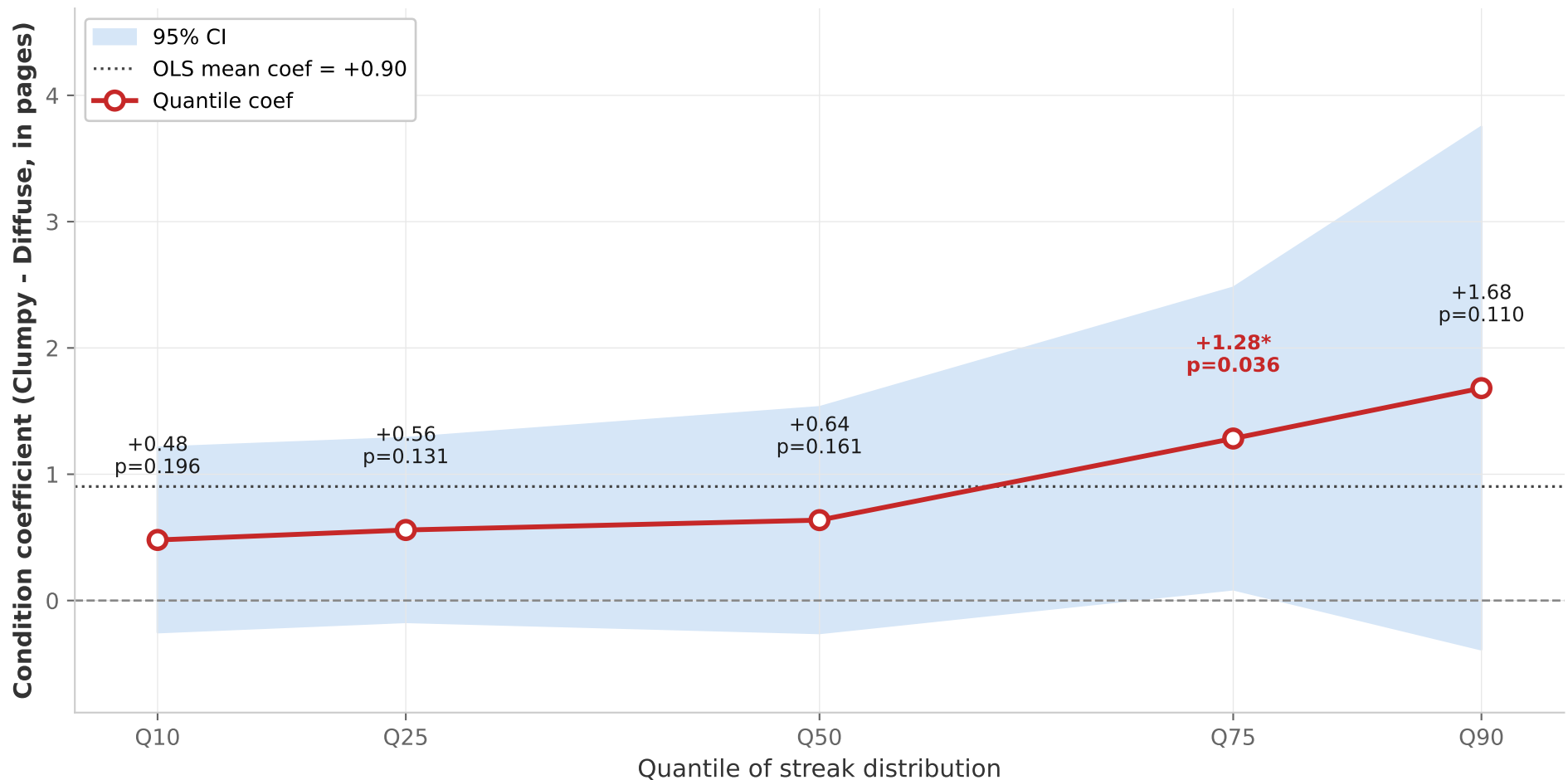
What the raincloud reveals beyond a bar+SEM chart

1. Both distributions are right-skewed (long tail of "deep readers"); mean is pulled above the median in each.
2. SD is 27% larger in Clumpy (2.35 vs 1.85) - the manipulation widens the spread, not just translates the centre.
3. The upper tail in Clumpy reaches further (max = 10.5 vs 9.75 pages, IQR top = 4.88 vs 4.00).

Hypothesis: the manipulation produces a heterogeneous treatment effect, concentrated in the upper tail. Tested formally on page 2.

Quantile regression: condition effect across the streak distribution

Model: Mean reading-only streak $\sim$ condition + final_answer_length (N = 99). Plotted: condition coefficient (Clumpy vs Diffuse).



How to read this

If the manipulation were a pure location shift, the coefficient line would be flat at the OLS mean (+0.90).

Instead the coefficient grows monotonically: $\sim +0.5$ at Q10/Q25, $+0.6$ at the median, $+1.3$ at Q75 (significant), $+1.7$ at Q90.

Only the Q75 coefficient is individually significant; sample size limits power at the extreme quantiles. The upward trend across quantiles is the s

n: the Clumpy condition does not move every participant equally - it primarily lengthens streaks among participants already in the up